2026 EDITION · SESSION 1 OF 13

Computational Biology

Life. Data. Models.

Bharuno Mahesworo, S.T., M.Sc.



This course bridges biology and computation

Education

Electrical Engineering · ITB
Biomedical Engineering · University of Glasgow

Experience

2018–21	Bioinformatics researcher · BINUS
2022	Data analyst · UNICEF
2022–23	Ministerial staff · Ministry of Health
2022–24	Lead data analyst · Salim Group EDTS
2024–now	AVP, Lead Data Analyst · DBS Bank



Bharuno “Uno” Mahesworo

Bioinformatics · statistics · analytics · applied AI

Learn the path from biological question to insight

Biological foundations come first; then sequences and databases become reproducible analysis with Python and Biopython.

01 **EXPLAIN**
Core biology + bioinformatics concepts

\+

02 **RETRIEVE**
Data from biological databases

\+

03 **COMPUTE**
Sequence analysis with Biopython

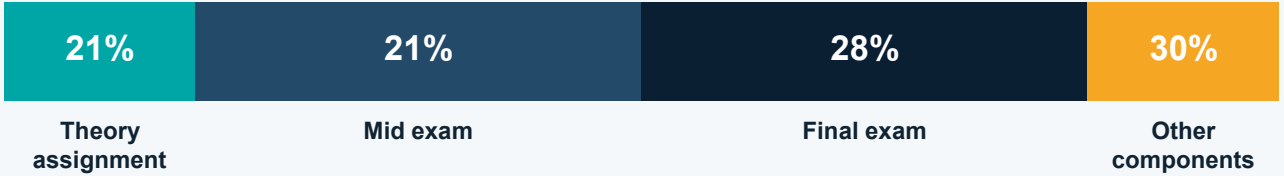
\+

04 **INTERPRET**
Results, assumptions and limitations

No prior programming is required; curiosity and verification are.

Lecture assessment carries 70% of the grade

70%
lecture evaluation



ASSESSMENT	WEIGHT	PRIMARY EVIDENCE
Theory assignment	**21%**	Concepts + reasoning
Mid exam	**21%**	Core foundations + application
Final exam	**28%**	Integrated computational analysis
Other components	**30%**	Confirm in the LMS / course outline

Thirteen sessions: molecules to computation

This is the learning arc—not a calendar. Confirm exact dates and assessment windows in the LMS.



Core references

Campbell Biology · Bioinformatics for Beginners · Current papers and databases introduced in class

01

SECTION 01

Life behaves as a system

Organization, information, energy and evolution interact across scales.

What is Biology?



Biology, study of living things and their vital processes. But life is not a single thing—it is a **process**. The field deals with all the physicochemical aspects of life.

Eight characteristics distinguish living systems

08

A living system shows the pattern as a whole. One trait alone does not define life.

01 **Order**
cellular organization

02 **Energy**
metabolism

03 **Regulation**
homeostasis

04 **Continuity**
growth and reproduction

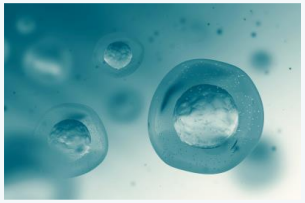
05 **Adaptation**
evolution across generations

06 **Response**
sensing the environment

07 **Inheritance**
genetic information

08 **Interaction**
relationships with other systems

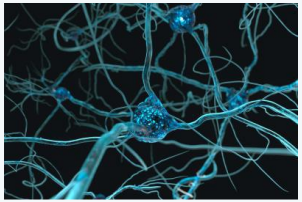
Unifying Themes in Biology



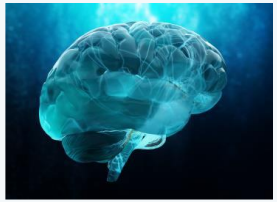
Cell theory describes the organization of living systems



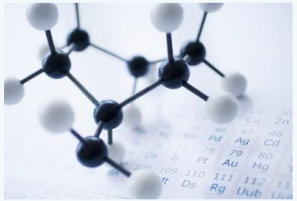
The molecular basis of inheritance explains the continuity of life



The relationship between structure and function underlies living systems



Evolutionary conservation explains the unity of living systems



Cells are information-processing systems



Living systems exist in a nonequilibrium state

The Hierarchy of Life

A cell is the smallest functional unit that can perform all of life's tasks. A living organism may consist of a single cell or a huge number of cells. In multicellular organisms, cells are specialized and depend on other cells to maintain life. The specialization and interdependence of cells contribute to a characteristic hierarchy of life, with each level defined by its structure and function.

Subatomic

Atom

Molecule

Organelle











CELL
Basic unit of life

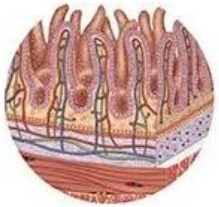
TISSUE
A group of specialized cells that work together for a particular function

ORGAN
A distinct structure made up of different tissues that have a specific function

ORGAN SYSTEM
A collection of organs that carry out specific functions within an organism

ORGANISM
A living thing that carries out all of life's functions











Population

Ecosystem

Biome

Biosphere

CAROLINA®
www.carolina.com

Biology needs both parts and systems

REDUCTIONISM

Isolate one component

Control variables

Map local mechanisms

Build a causal explanation

Deep mechanism

SYSTEMS THINKING

Integrate interacting components

Model feedback and nonlinearity

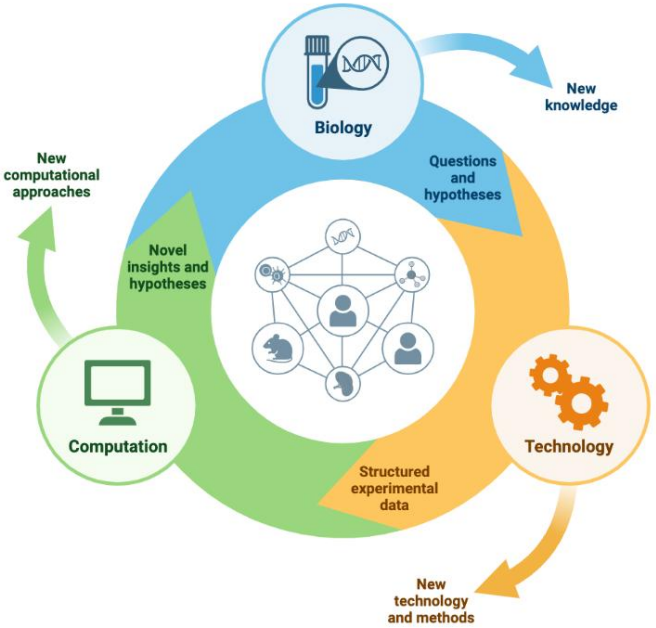
Connect scales

Predict collective behavior

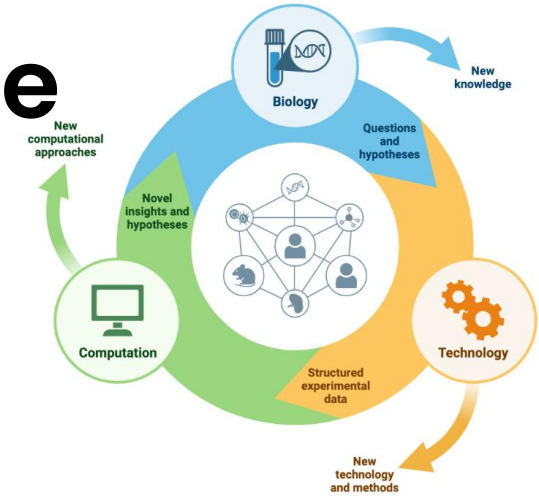
Whole-system context

Systems Biology

Systems biology is an interdisciplinary field that studies complex living systems as a whole, focusing on the interactions between components (like genes, proteins, cells) rather than just individual parts, using computational models and large-scale data to understand how these interactions create functions and behaviors, moving beyond reductionism to a holistic view.



Systems Biology Example



1. Biology → Question
People feel **hungry, tired, and unfocused** after not eating for several hours.
Question:
How does the body sense low energy and trigger hunger?

2. Technology → Data Collection
Researchers measure:
- Blood glucose levels over time
- Hormone levels (insulin, glucagon, ghrelin)
- Gene expression in liver and brain cells

Data from blood tests, molecular assays, and clinical measurements

3. Computation → Data Analysis
Computational models:
- Analyze correlations between glucose and hormone levels
- Model feedback loops controlling hunger
- Identify regulatory pathways responding to low energy

Falling glucose triggers hormone signaling cascades
Multiple organs coordinate the response

4. Computation → New Hypothesis
Hunger is controlled by a **distributed regulatory network**, not a single “hunger switch.”

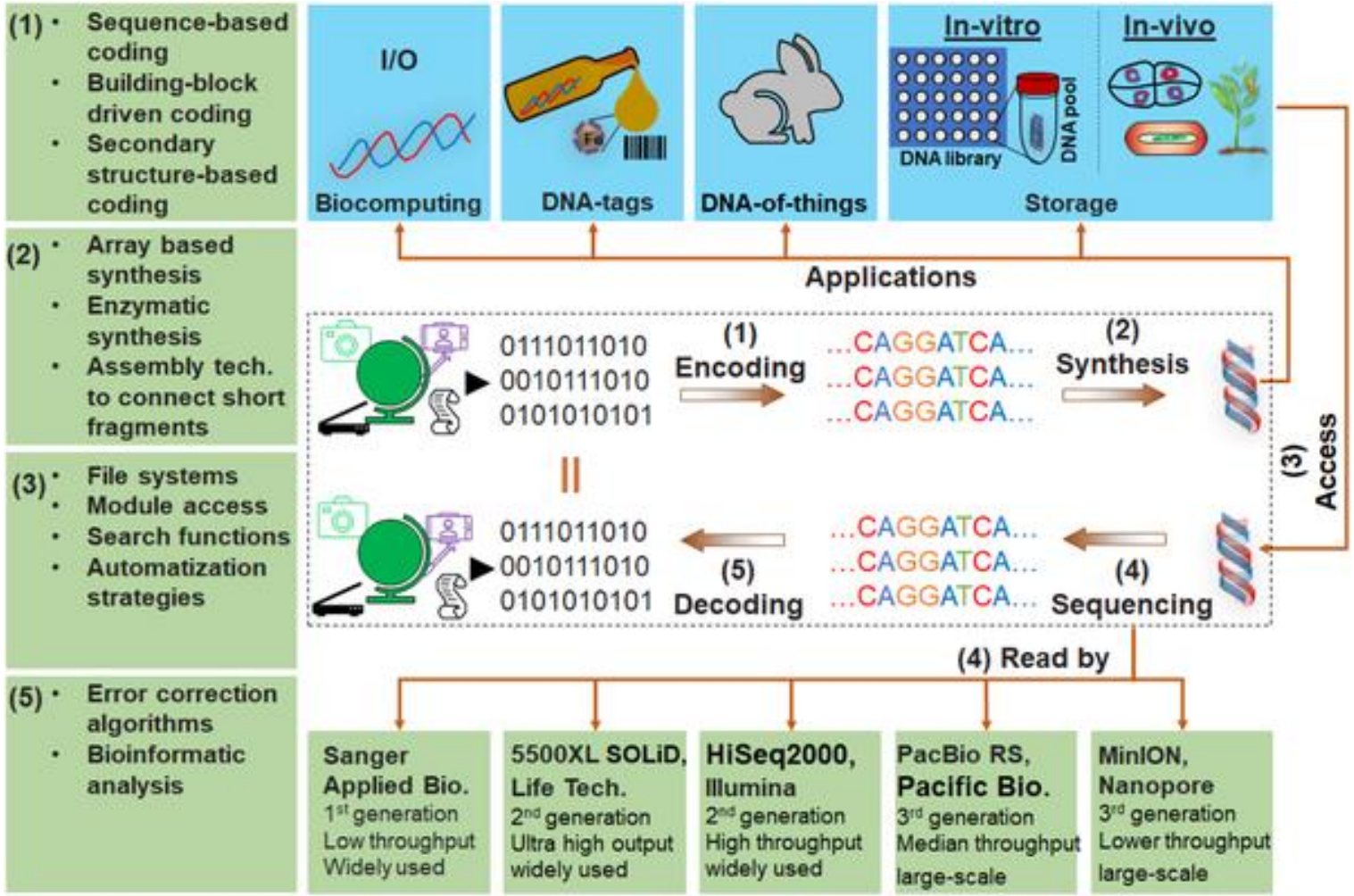
5. Biology → New Knowledge
Explains why hunger feels physical *and* hormonal
Guides interventions for diabetes and obesity
Leads to better dietary recommendations
Cycle restarts with refined models

Why Do we need ‘computer’?

Because life is a complex, data-rich system.

- **Scale**: Biological systems involve thousands of interacting components across multiple levels (DNA, RNA, cells, organs).
- **Data volume**: Modern biology produces millions to billions of data points (genomes, expression profiles, omics).
- **Interactions matter**: Function emerges from networks and feedback loops, not isolated molecules.
- **Non-linearity**: Small molecular changes can lead to large, system-wide effects.
- **Limits of experiments**: Wet-lab experiments are costly, slow, and partial; not all hypotheses can be tested directly.
- **Role of computation**: Store and organize massive datasets, discover patterns and relationships, integrate multiple data types, and build predictive and simulation models

In systems biology, computers are not optional tools—they are essential partners for understanding life as a system.


(1)
Sequence-based coding
Building-block driven coding
Secondary structure-based coding
(2)
Array based synthesis
Enzymatic synthesis
Assembly tech. to connect short fragments
(3)
File systems
Module access
Search functions
Automatization strategies
(5)
Error correction algorithms
Bioinformatic analysis
I/O
Biocomputing
DNA-tags
DNA-of-things
In-vitro
In-vivo
DNA library
DNA pool
Storage
Applications
0111011010
0010111010
0101010101
(1) Encoding
...CAGGATCA...
...CAGGATCA...
...CAGGATCA...
(2) Synthesis
(3) Access
(4) Sequencing
(5) Decoding
(4) Read by
Sanger Applied Bio.
1st generation
Low throughput
Widely used
5500XL SOLiD, Life Tech.
2nd generation
Ultra high output
widely used
HiSeq2000, Illumina
2nd generation
High throughput
widely used
PacBio RS, Pacific Bio.
3rd generation
Median throughput
large-scale
MinION, Nanopore
3rd generation
Lower throughput
large-scale

Recent Major Advancement in System Biology

nature

Explore content | About the journal | Publish with us | Subscribe

nature > news > article

NEWS | 11 November 2024 | Correction 14 November 2024

AI protein-prediction tool AlphaFold3 is now more open

The code underlying the Nobel-prize-winning tool for modelling protein structures can now be downloaded by academics.

nature

Explore content | About the journal | Publish with us | Subscribe

nature > news > article

NEWS | 22 March 2024 | Clarification 22 March 2024 | Correction 22 March 2024 | Update 28 March 2024 | Correction 28 March 2024

First pig kidney transplant in a person: what it means for the future

The operation's early success has made researchers hopeful that clinical trials for xenotransplanted organs will start soon.

By Smriti Mallapaty & Max Kozlov